

Sahilkumar Prasad

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Professional Summary

Backend Engineer with **3+ years of experience** designing scalable backend systems using **Java and Spring Boot**. Experienced in building microservices, REST APIs, and distributed systems across **Agile/Lean delivery environments**. Strong expertise in **Kafka-based event-driven architecture, Redis caching, and database optimization**. Proven ability to write clean, maintainable code with a commitment to **TDD, CI/CD, and continuous improvement**. Thrives in collaborative, transparent teams and is passionate about learning, knowledge sharing, and building the right solutions for real business problems.

Technical Skills

Languages: Java, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs

Architecture: Microservices, Distributed Systems, Event-Driven Architecture, System Design (LLD, HLD), OOP

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Messaging Systems: Apache Kafka

Cloud & Infrastructure: AWS (EC2, S3, RDS), Docker, Kubernetes

Testing: JUnit, Mockito, Test-Driven Development (TDD)

Engineering Practices: Agile, Lean, Continuous Integration, Continuous Delivery, GitHub Actions, CI/CD Pipelines

Tools: Git, GitHub Actions, Docker

Professional Experience

Software Development Engineer

May 2023 – Present

Crescent Engineering and Consulting, Navi Mumbai

- Architected and deployed **8+ Spring Boot microservices** supporting core business workflows, handling **500K+ requests/day** with **99.9% uptime** on cloud infrastructure using **AWS and Docker**.
- Reduced **API latency by 45%** by implementing Redis caching and asynchronous request processing, directly improving end-user experience and SLA compliance.
- Improved **database query performance by 50%** through targeted indexing strategies, query optimization, and connection pool tuning across PostgreSQL and MySQL environments.
- Migrated tightly coupled services into a **Kafka-based event-driven architecture**, enabling reliable asynchronous processing and decoupling across multiple services.
- Implemented secure authentication and authorization using **Spring Security, JWT, and RBAC**, ensuring compliance with enterprise security policies across all microservices.
- Practised **Test-Driven Development (TDD)** using JUnit and Mockito, achieving high test coverage and reducing post-deployment defects; delivered features iteratively within an **Agile/Scrum** team.
- Built and maintained **CI/CD pipelines using GitHub Actions**, automating build, test, and deployment stages to ensure fast, reliable releases to staging and production environments.

Key Projects

Distributed Notification System

github.com/Sahill1001/notification-system

- Built a scalable notification platform (email, SMS, in-app) using **Kafka-based event-driven architecture** with asynchronous consumers, dead-letter queues, and Redis-based rate limiting.

Distributed Rate Limiter

github.com/Sahill1001/rate-limiter

- Implemented a distributed rate limiter using **Spring Boot and Redis** (Token Bucket algorithm) achieving **50K+ requests/minute** with cross-node consistency via atomic Redis operations.

Distributed URL Shortener

github.com/Sahill1001/url-shortener

- Developed a high-performance URL shortener using **Spring Boot, PostgreSQL, Redis, and Kafka**, reducing read latency by **80%** via caching and processing analytics asynchronously.

Education

University of Mumbai

Bachelor of Science in Computer Science

2020 – 2023

CGPA: **9.57 / 10**

Achievements

- Solved **300+ Data Structures and Algorithms problems** on LeetCode, demonstrating strong Object-Oriented and algorithmic problem-solving skills.
- Designed and operated backend systems handling **500K+ production requests daily** with 99.9% uptime on cloud infrastructure.
- Built distributed backend projects demonstrating real-world application of **caching, rate limiting, event-driven architecture, and CI/CD best practices**.